SCDL3991: Regret Bound Analysis of Multi-Armed Bandit Problems

Author: Robert Zhang
SID: 540858851

THE UNIVERSITY OF SYDNEY

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1 Introduction

1.1 Background

Multi-armed bandits is a simple yet powerful framework of algorithms that make sequential decisions under uncertainty. The name originates from the problem of choosing between several slot machines ("one-armed bandits"), each with an unknown reward distribution. At each time step, an agent selects one of the available actions, known as set of arms and observes only the reward of the chosen arm. The goal is to maximise the total cumulative rewards over a finite time horizon by identifying and favouring high-reward arms while continuing to learn about uncertain alternatives. This problem setting captures the fundamental trade-off between exploration and exploitation: the need to learn about uncertain arms while also being able to simultaneously exploit currently known information.

The concept of multi-armed bandits has been studied since the 1930s, but its formal mathematical model was not established until the 1950s, initially applied to clinical trials by the Center for Drug Evaluation and Research with the aim of identifying the most effective treatment [1]. Many real-world systems require making sequential decisions under uncertainty such as: online advertising, clinical trials, recommender systems etc [2]. In these settings, the decision-maker does not have prior knowledge of the reward distributions associated with each action, and must learn these distributions through interaction over time. Feedback is only available for actions that are taken, meaning that information gathered is inherently partial and must be gathered sequentially. As a result, naive strategies that focus only on immediate reward often perform poorly in the long run, while purely exploratory strategies are inefficient and costly. This highlights the need for systematic strategies that balance exploring less-known actions and exploiting currently estimated best actions. The multi-armed bandit framework provides a formal way to study this trade-off.

1.2 Aims

The aim of this research project is to analyse and compare different classical bandit algorithms in terms of their ability to manage the fundamental exploration-exploitation tradeoff. This report will focus on both theoretical performance guarantees and empirical behaviour, with particular attention to regret as a measure of performance over time.

1.3 Roadmap

This report is primarily based on the Introduction to Multi-Armed Bandits by Microsoft Research and follows its presentation of several classical bandit algorithms and their theoretical guarantees. Where appropriate, additional explanations and derivations are provided to develop a deeper understanding of the underlying concepts.

The remainder of this report is organised as follows. Section 2 introduces the formal multi-armed bandit framework and establishes the notation used throughout the report. We define the stochastic bandit setting, reward distributions, expected rewards, optimal arms, and cumulative regret, which serves as the primary performance metric for evaluating bandit algorithms.

Section 3 presents several fundamental algorithms for balancing exploration and exploitation. In particular, we study Uniform Exploration, Successive Elimination, and

Thompson Sampling. For each algorithm, we describe its decision-making mechanism, and analyse its theoretical regret guarantees by deriving upper bounds on its performance.

Section 4 examines the fundamental limitations of the multi-armed bandit problem. We present a regret lower bound that applies to all bandit algorithms, demonstrating that no strategy can avoid a certain amount of exploration and establishing a benchmark against which algorithmic performance can be compared.

Finally, Section 5 provides an empirical evaluation of the algorithms through a numerical example. We compare the observed performance with the theoretical regret bounds developed earlier, highlighting the practical trade-offs between exploration and exploitation and illustrating how theoretical guarantees manifest in finite-horizon settings. The report concludes with a summary of the main findings and a discussion of future research into multi-armed bandits.

2 Problem Formulation and Setup

Consider a stochastic multi-armed bandit problem with K possible arms which are indexed from $1, 2, \cdots K$, and T rounds that we play over. Each arm a is associated with an unknown reward distribution denoted by $\mathcal{D}_a$ with mean reward μ_a where $\mu(a) = \mathbb{E}[\mathcal{D}_a]$. For each round $t = 1, 2, \ldots, T$ a bandit algorithm only observes and collects a reward for the selected arm. The algorithm's goal is to maximise its total reward over the T rounds.

We make three key assumptions about this model: the algorithm observes only the reward for the selected action, and nothing else. In particular, it does not observe rewards for other actions that could have been selected. The reward for each arm is assumed to be independent and identically distributed (i.i.d.). For each arm a that we choose, we independently sample a reward from its reward distribution $\mathcal{D}_a$ which is unknown to the algorithm. Per-round rewards are bounded; we will enforce the restriction to the interval $[0, 1]$ for simplicity.

Thus a bandit algorithm is summarised as:

Algorithm 1 Stochastic Bandits

Parameters: K arms, T rounds (known), reward distribution $\mathcal{D}_a$ for each arm a (unknown)

for $t = 1$ **to** T **do**

  Choose arm a_t,

  Reward $r_t \in [0, 1]$ is sampled independently from distribution $\mathcal{D}_a$, $a = a_t$

  Algorithm collects r_t and observes nothing else

end for

In this algorithm and subsequent bandit algorithms, we are interested in the mean reward vector $\mu \in [0, 1]^K$, where $\mu(a) = \mathbb{E}[\mathcal{D}_a]$. For simplicity we will use the Bernoulli distribution to model rewards, when the reward of each reward a can be either 1 or 0 ("success or failure"). The problem instance is specified by the time horizon T and the mean reward vector. The best mean reward is $\mu^* := \max_{a \in A} \mu(a)$. The difference $\Delta(a) = \mu^* - \mu(a)$ describes how bad arm a is compared to μ^* ; we call the gap of arm a. An optimal arm is an arm a with $\mu(a) = \mu^*$. We measure performance of an algorithm by comparing the cumulative reward to the best-arm benchmark $\mu^* \cdot T$: where Regret $R(T)$ is defined as:

$$R(T) = \mu^* T - \sum_{t=1}^{T} \mu_{a_t} \tag{2.1}$$

Note that the selected arm a_t itself is random variable, since the algorithm's choice can depend on previously observed random rewards and may also involve randomised decision rules. Consequently, the cumulative regret $R(T)$ is also a random variable. We analyse the performance of an algorithm with regard to the dependence on the number of arms K and the time horizon T and use big-O notation to focus on the asymptotic dependence on the parameters of interest, rather than keep track of the constants.

3 Methodology

3.1 Uniform Exploration

The first multi-armed bandit algorithm that we will analyse is uniform exploration. The idea behind this algorithm is that we explore all arms uniformly (at the same rate), regardless of what has been observed previously, and then pick the empirically best arm for exploitation. This leads to the explore-first algorithm where we dedicate an initial amount of rounds for exploration and then the rest for exploration. This algorithm can be summarised as follows:

Algorithm 2 Explore-First with parameter N

1: **Exploration phase:** try each arm N times
2: Select the arm $\hat{a}$ with the highest average reward (break ties arbitrarily)
3: **Exploitation phase:** play arm $\hat{a}$ in all remaining rounds

The parameter N is fixed and will be determined as a function of K and T so as to minimise regret. Let us now analyse the expected regret of this algorithm. Let the average reward for each action a after exploration phase be denoted $\mu(\bar{a})$. We define $\mu(\bar{a}) := \frac{1}{N}\sum_{i=1}^{N} X_i(a)$ where $X_i(a)$ is the random variable denoting the reward you observe on the i-th pull of arm a. We have that $X_i(a) \overset{\text{i.i.d.}}{\sim} \mathcal{D}_a$ for $i = 1, \dots, N$ as each pull of arm a returns an independent sample from the reward distribution $\mathcal{D}_a$. We want this empirical average reward $\mu(\bar{a})$ to be a good estimate of the true estimate rewards for each arm meaning the quantity $|\mu(\bar{a}) - \mu(a)|$ should be small. We bound it using the Hoeffding inequality [3] - if $X_1, \dots, X_n$ are independent random variables taking values in $[0, 1]$ with mean μ, and $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, then for every $\varepsilon > 0$,

$$\Pr\big(|\bar{X} - \mu| \geq \varepsilon\big) \leq 2e^{-n\varepsilon^2/2}.$$

Applying this inequality to the N independent rewards obtained from arm a, we obtain

$$\Pr[|\bar{\mu}(a) - \mu(a)| \leq \text{rad}] \geq 1 - \frac{2}{T^4}, \text{ where } \text{rad} := \sqrt{\frac{2\log T}{N}} \tag{3.1}$$

Thus, $\mu(a)$ lies in the confidence interval $[\,\bar{\mu}(a) - \text{rad},\ \bar{\mu}(a) + \text{rad}\,]$ with high probability and rad is the confidence radius of this interval. We define the clean event as the event that the deviation bound in (3.1) holds for all arms simultaneously, i.e. $\mathcal{E} := \bigcap_{a=1}^{K} \{|\hat{\mu}(a) - \mu(a)| \leq \text{rad}\}$. Using this we can ignore low-probability "bad events" in the analysis. This simplifies proofs, at the cost of slightly larger constants in asymptotic bounds.

Theorem 3.1. *Explore-first achieves regret* $\mathbb{E}[R(T)] = O\Big(T^{2/3}\,(K\log T)^{1/3}\Big)$ *by choosing N optimally*

Proof. for K arms we want all arms to simultaneously satisfy $\forall a \in [K] : |\bar{\mu}(a) - \mu(a)| \leq \text{rad}$, and by the union bound we have $\Pr[\forall a \in [K] : |\bar{\mu}(a) - \mu(a)| \leq \text{rad}] \geq 1 - \frac{2K}{T^4}$. Each arm is chosen N times and each round in exploration contributes trivially at most 1 to regret. For the exploitation phase, let the best arm be $a^*$and suppose the algorithm chooses an arm $a \neq a*$. This must have been because its average reward was better than that of a^*: $\bar{\mu}(a) > \bar{\mu}(a^*)$. Since this is a clean event we have:

$$\mu(a) + \text{rad} \geq \bar{\mu}(a) > \bar{\mu}(a^*) \geq \mu(a^*) - \text{rad}.$$

Re-arranging the terms, it follows that $\mu(a^*) - \mu(a) \leq 2\text{rad}$. Thus, each round in the exploration phase contributes 2rad to regret. We combine these contributions to get the following:

$$R(T) \leq KN + 2\text{rad} \cdot T$$

We then choose a N that balances exploration regret KN and exploitation regret. We know the per round regret in the exploitation phase is upper-bounded by 2rad. This means we can express the regret of the exploration phase in terms of big-O notation as follows:

$$\text{regret per round} \leq 2rad = 2 \cdot \sqrt{\frac{2\log T}{N}}$$

In the worst case, this affects up to T rounds and this gives us a big-O bound for the regret of $R_{exploit}$:

$$O(T\text{rad}) = R_{exploit} \leq 2T\sqrt{\frac{2\log T}{N}}$$

We can now express the total regret as a big-O asymptotic function (after dropping constants)

$$R(T) = O(KN) + O(T\text{rad}) = O(KN + T\sqrt{\frac{\log T}{N}})$$

We can now balance the two summands inside the asymptotic formula making exploration regret $\approx$ exploitation regret and rearrange for N to get:

$$KN = T\sqrt{\frac{\log T}{N}}$$

Solving for N:

$$KN^{3/2} = T\sqrt{\log T}$$

$$N = \left(\frac{T}{K}\right)^{2/3} (\log T)^{1/3}$$

Finally, substitute $N = \left(\frac{T}{K}\right)^{2/3} (\log T)^{1/3}$ to obtain:

$$\mathbb{E}[R(T)] = O\Big(T^{2/3} (K \log T)^{1/3}\Big) \tag{3.2}$$

The probability of the bad event is at most $2/T^4$. Since the regret in any round is bounded by 1, the total regret contributed by the bad event is at most

$$T \cdot \frac{2}{T^4} = \frac{2}{T^3},$$

which is negligible. Therefore the expected regret is dominated by the regret on the good event, yielding

$$\mathbb{E}[R(T)] = O\Big(T^{2/3}(K \log T)^{1/3}\Big).$$

$\square$

3.2 Epsilon-Greedy

A problem with the explore-first approach is that its performance can be very bad if many arms have a large gap $\Delta(a)$. A better way is to spread the exploration more uniformly over time. This is done by the epsilon greedy algorithm.

Algorithm 3 Epsilon-Greedy Algorithm

```
Input: K arms, time horizon T, exploration probabilities {ε_t}_{t=1}^T
for t = 1 to T do
    Toss a coin with success probability ε_t
    if coin shows success then
        Explore: choose an arm uniformly at random
    else
        Exploit: choose the arm with the highest average reward so far
    end if
    Pull the chosen arm, observe reward, and update the average reward
end for
```

Theorem 3.2. *The ε-greedy algorithm with exploration probability $\varepsilon_t = t^{-1/3}(K \log t)^{1/3}$ satisfies $\mathbb{E}[R(T)] = O\big(T^{2/3}(K \log T)^{1/3}\big)$.*

Proof. Fix a round t. During exploration, an arm is chosen uniformly at random with probability ε_t. Therefore, by round t, each arm has been explored on average $\frac{t\varepsilon_t}{K}$ times. Applying Hoeffding's inequality, for any arm a and any $\delta > 0$,

$$\Pr(|\hat{\mu}_a - \mu_a| \geq \delta) \leq 2e^{-\frac{2t\varepsilon_t\delta^2}{K}}.$$

Choosing

$$\delta = \sqrt{\frac{2K \log t}{t\varepsilon_t}},$$

gives

$$\Pr\left(|\hat{\mu}_a - \mu_a| \geq \sqrt{\frac{2K \log t}{t\varepsilon_t}}\right) \leq \frac{2}{t^4}.$$

Hence, with high probability, the empirical mean of every arm is within $\sqrt{\frac{2K \log t}{t\varepsilon_t}}$ of its true mean. Then using a similar argument as in the analysis of uniform exploration gives a regret at round t of at most

$$\hat{\mu}(a^*) - \mu(a_t) \leq 2\sqrt{\frac{2K \log t}{t\varepsilon_t}}.$$

The expected regret incurred at round t is bounded by

$$\mathbb{E}[\widetilde{R}(t)] \leq \varepsilon_t + (1 - \varepsilon_t)2\sqrt{\frac{2K \log t}{t\varepsilon_t}} \leq \varepsilon_t + 2\sqrt{\frac{2K \log t}{t\varepsilon_t}}$$

Balancing the two terms yields $\varepsilon_t = t^{-1/3}(K \log t)^{1/3}$ and then substituting this choice gives $\mathbb{E}[R(t)] = O\big((K \log t)^{1/3} t^{-1/3}\big)$ regret per round.

Summing over all rounds,

$$\mathbb{E}[R(T)] = \sum_{t=1}^{T} \mathbb{E}[R(t)) = O\left((K \log T)^{1/3} \sum_{t=1}^{T} t^{-1/3} \right).$$

Since

$$\sum_{t=1}^{T} t^{-1/3} = O(T^{2/3}),$$

we obtain

$$\mathbb{E}[R(T)] = O\big(T^{2/3}(K \log T)^{1/3}\big)\,.$$

□

3.3 Successive Elimination

The next multi-armed bandit algorithm that we will analyse is successive elimination. This algorithm adapts exploration to the observations so that very under-performing arms are phased out sooner. In order to do this let us introduce the idea of confidence bounds.

Fix a round t and an arm a, and let $n_t(a)$ be the number of rounds before t in which arm a is chosen, and $\bar{\mu}_t(a)$ be the average reward in those rounds. We again use the Hoeffding inequality to derive:

$$\Pr(|\bar{\mu}_t(a) - \mu(a)| \leq r_t(a)) \geq 1 - \frac{2}{T^4}, \qquad r_t(a) = \sqrt{\frac{2 \log T}{n_t(a)}}. \tag{3.3}$$

Since $n_t(a)$ is random, we introduce a reward tape of i.i.d. samples from $\mathcal{D}_a$ for each arm a. When $n_t(a) = j$, the empirical mean of arm a equals $\bar{v}_j(a)$, the average of the first j tape entries. Hence, for every fixed j, Hoeffding's inequality gives:

$$\Pr(|\bar{v}_j(a) - \mu(a)| \leq r_t(a)) \geq 1 - \frac{2}{T^4}.$$

Taking a union bound, it follows (assuming K, the number of arms, satisfies $K \leq T$) that

$$\Pr[E] \geq 1 - \frac{2}{T^2}, \qquad E := \{\forall a\, \forall t\ \ |\bar{\mu}_t(a) - \mu(a)| \leq r_t(a)\}\,. \tag{3.4}$$

The event E in equation (3.4) will be the clean event for the subsequent analysis. For each arm a at round t, we define the upper and lower confidence bounds

$$\mathrm{UCB}_t(a) = \bar{\mu}_t(a) + r_t(a), \qquad \mathrm{LCB}_t(a) = \bar{\mu}_t(a) - r_t(a).$$

The confidence interval is $\big[\mathrm{LCB}_t(a), \mathrm{UCB}_t(a)\big]$ with confidence radius $r_t(a)$.

Algorithm 4 Successive Elimination for K Arms

Input: K arms, horizon T
All arms are initially designated as active
while there is more than one active arm **do**
  new phase: play each active arm once
  for each active arm a **do**
    if $\mathrm{UCB}_t(a) < \mathrm{LCB}_t(a')$ for some other active arm a' **then**
      Deactivate arm a {deactivation rule}
    end if
  end for
end while
Play the remaining active arm forever

Theorem 3.3. *The Successive Elimination algorithm achieves* $\mathbb{E}[R(t)] = O(\sqrt{Kt\log T})$ *for all rounds* $t \leq T$.

Proof. Let a^* be an optimal arm which cannot be deactivated. Fix any arm a such that $\mu(a) < \mu(a^*)$. Consider the last round $t \leq T$ when the deactivation rule was invoked and arm a remained active. The confidence intervals of a and a^* must overlap at round t. Therefore,

$$\Delta(a) := \mu(a^*) - \mu(a) \leq 2\big(r_t(a^*) + r_t(a)\big) = 4\,r_t(a)$$

The last equality is because $n_t(a) = n_t(a^*)$, since the algorithm has been alternating active arms and both a and a^* have been active before round t. Thus, we have the following crucial property:

$$\Delta(a) \leq O\big(r_T(a)\big) = O\Big(\sqrt{\frac{\log(T)}{n_T(a)}}\Big) \quad \text{for each arm } a \text{ with } \mu(a) < \mu(a^*). \tag{3.5}$$

The contribution of arm a to regret at round t, denoted by $R(t;a)$ can be expressed as $\Delta(a)$ for each round this arm is played:

$$R(t;a) = n_t(a)\cdot\Delta(a) \leq n_t(a)\cdot O\Big(\sqrt{\frac{\log T}{n_t(a)}}\Big) = O\Big(\sqrt{n_t(a)\log T}\Big).$$

Summing up over all arms, we obtain that:

$$R(t) = \sum_{a\in A} R(t;a) \leq O\Big(\sqrt{\log T}\sum_{a\in A}\sqrt{n_t(a)}\Big). \tag{3.6}$$

Since $f(x) = \sqrt{x}$ is concave and $\sum_{a\in A} n_t(a) = t$, by Jensen's inequality we have

$$\frac{1}{K}\sum_{a\in A}\sqrt{n_t(a)} \leq \sqrt{\frac{1}{K}\sum_{a\in A} n_t(a)} = \sqrt{\frac{t}{K}}.$$

Plugging into 3.6

$$R(t) \leq O\Big(\sqrt{Kt\log T}\Big).$$

□

Another approach for adaptive exploration is optimism under uncertainty: we assume each arm is as good as it can possibly be given the observations so far, and choose the best arm based on these optimistic estimates. This leads to an algorithm called *UCB*:

Algorithm 5 UCB Algorithm

Try each arm once
for $t = 1, \ldots, T$ **do**
    Pick an arm a that maximizes $\mathrm{UCB}_t(a)$
end for

Theorem 3.4. *The UCB algorithm achieves the same* $\mathbb{E}[R(t)] = O(\sqrt{Kt \log T})$ *for all rounds* $t \leq T$ *as successive elimination.*

The proof of this is very similar to the proof for deriving the regret bound for successive elimination.

3.4 Thompson Sampling

The final multi-armed bandit algorithm that we will analyse is Thompson Sampling. This model assumes that the problem instance I is drawn from a known prior distribution $\mathbb{P}$ over mean reward vectors $\mu \in [0,1]^K$. Conditional on μ, each arm a yields rewards from a distribution $D_{\mu(a)}$, where $(D_\nu)_{\nu \in [0,1]}$ is a known single-parameter family satisfying $\mathbb{E}[D_\nu] = \nu$. Hence, each bandit instance is fully characterised by the mean vector μ.

The goal is to minimise Bayesian regret:

$$BR(T) = \mathbb{E}_{\mu \sim \mathbb{P}}\Big[\mu^* T - \sum_{t=1}^{T} \mu(a_t)\Big] \tag{3.7}$$

where $\mu^* = \max_{a \in [K]} \mu(a)$.

At each round t, the algorithm observes the history

$$H_t = ((a_1, r_1), \ldots, (a_{t-1}, r_{t-1})),$$

and maintains a posterior belief $\mathbb{P}_H(\mu) = \Pr(\mu \mid H_t = H)$ over the unknown mean vector. This posterior summarises everything the algorithm has learned about μ from the rewards observed so far.

By Bayes' rule, the posterior depends only on the observed history, not on which algorithm produced it. When arm a is pulled and reward r is observed, the posterior updates as

$$\mathbb{P}_{H \oplus (a,r)}(\mu) \propto \mathbb{P}_H(\mu)\, D_{\mu(a)}(r),$$

where $H \oplus (a, r)$ denotes the history H extended by the new observation (a, r). This says the new posterior is proportional to the old posterior weighted by the likelihood of observing reward r from arm a with mean $\mu(a)$. Crucially, this update can be applied incrementally as new observations arrive.

Here $\mathbb{P}(\mu)$ denotes the prior probability - the initial joint belief over the full mean reward vector μ before any rewards are observed of the mean vector μ, - and $\mathbb{P}^a$ denotes the marginal prior on arm a's mean $\mu(a)$ alone. Independence across arms means the prior belief about each arm's mean is unaffected by the others.

$$\mathbb{P}(\mu) = \prod_{a \in [K]} \mathbb{P}^a(\mu(a)),$$

then the posterior factorises:

$$\mathbb{P}_H(\mu) = \prod_{a \in [K]} \mathbb{P}_H^a(\mu(a)).$$

Hence, each arm can be updated independently using only its own observations.

Thompson Sampling selects arms according to their posterior probability of being optimal:

$$p_t(a) = \Pr(a^* = a \mid H_{t-1}).$$

Thompson Sampling can be implemented by sampling from the posterior distribution over mean reward vectors. Equivalently, the algorithm can be written as:

Algorithm 6 Thompson Sampling (posterior sampling form)

```
1: for each round t = 1, 2, ..., T do
2:     Observe history H_{t-1}
3:     Sample μ_t ~ P(μ | H_{t-1})
4:     Pull arm a_t = arg max_{a∈[K]} μ_t(a)
5: end for
```

These two formulations are equivalent in distribution:

$$\Pr(a_t = a \mid H_{t-1}) = \Pr(a^* = a \mid H_{t-1}),$$

so sampling from the posterior over mean vectors induces the same arm-selection probabilities as sampling according to posterior optimality. This interpretation shows that Thompson Sampling acts by sampling a plausible environment from the posterior and acting optimally under it.

We will now analyse the expected regret bound of Thompson Sampling.

Theorem 3.5. *The Bayesian regret of Thompson Sampling satisfies:* $BR(T) = O\left(\sqrt{KT \log T}\right)$

Proof. for each arm a and round t, define confidence bounds as in successive elimination:

$$\begin{aligned} r_t(a) &= \sqrt{\frac{2 \log T}{n_t(a)}}, \\ U_t(a) &= \bar{\mu}_t(a) + r_t(a), \\ L_t(a) &= \bar{\mu}_t(a) - r_t(a), \end{aligned} \tag{3.8}$$

and let $r(a, H_t) = (U_t(a) - L_t(a))/2$.
More generally, assume upper and lower confidence bounds $U(a, H_t), L(a, H_t)$ satisfy for some constant $\gamma > 0$:

$$\mathbb{E}\left[(U(a, H_t) - \mu(a))^-\right] \le \frac{\gamma}{TK}, \qquad \mathbb{E}\left[(\mu(a) - L(a, H_t))^-\right] \le \frac{\gamma}{TK}. \tag{3.9}$$

The key step is to bound the Bayesian regret in terms of the widths of the confidence intervals. Conditioned on the history H_t, Thompson Sampling has the property that a_t and the optimal arm a^* are identically distributed under the posterior. Therefore,

$$\mu(a^*) - \mu(a_t) = (U_t(a_t) - \mu(a_t)) + (\mu(a^*) - U_t(a^*)) + (U_t(a^*) - U_t(a_t)).$$

Since a_t and a^* have the same posterior distribution, the final term has expectation zero. Taking expectations and applying (3.9) to the remaining terms yields

$$BR(T) \leq 2\gamma + 2\sum_{t=1}^{T} \mathbb{E}[r(a_t, H_t)].$$

Using (3.9) with $\gamma = 2$,

$$BR(T) \lesssim \sum_{t=1}^{T} \mathbb{E}\left[\frac{1}{\sqrt{n_t(a_t)}}\right].$$

Rewriting by arms,

$$\sum_{t=1}^{T} \frac{1}{\sqrt{n_t(a_t)}} = \sum_{a\in[K]} \sum_{j=1}^{n_T(a)} \frac{1}{\sqrt{j}} = \sum_{a\in[K]} O(\sqrt{n_T(a)}).$$

Hence,

$$BR(T) \lesssim \sum_{a\in[K]} \sqrt{n_T(a)}.$$

Applying the Cauchy–Schwarz inequality, and using $\sum_a n_T(a) = T$,

$$\sum_{a\in[K]} \sqrt{n_T(a)} \leq \sqrt{KT}.$$

Therefore,

$$BR(T) = O\left(\sqrt{KT\log T}\right).$$

□

We can now summarise and compare the theoretical regret guarantees of the algorithms analysed so far.

Algorithm	**Theoretical Regret Bound**
Uniform Exploration	$O\left(T^{2/3}\,(K\log T)^{1/3}\right)$
Greedy-Epsilon	$O\left(T^{2/3}\,(K\log T)^{1/3}\right)$
Successive Elimination	$O(\sqrt{KT\log T})$
UCB	$O(\sqrt{KT\log T})$
Thompson Sampling	$O(\sqrt{KT\log T})$

Table 1: **Comparison of regret bounds and optimality of bandit algorithms.**

4 Algorithm Optimality and Lower Bounds

We now turn to the lower bound analysis. In particular, we show that no algorithm can achieve regret smaller than order $O(\sqrt{KT})$ on all problem instances. s lower bound establishes a fundamental limit on performance and demonstrates that the exploration-exploitation trade-off is unavoidable.

Theorem 4.1. *Fix the time horizon T and the number of arms K. Fix a bandit algorithm. Choose an arm a uniformly at random, and run the algorithm on problem instance I_a. Then the expected regret satisfies*

$$\mathbb{E}[R(T)] \geq O(\sqrt{KT}),$$

where the expectation is over the choice of arm a, the randomness in rewards, and any randomness in the algorithm.

4.1 Proof of Fundamental Lower Bound

4.1.1 KL Divergence

The key idea is that the ability of an algorithm to distinguish between similar instances is fundamentally limited by the amount of information it gathers. We formalise this using Kullback-Leibler (KL) divergence, For two probability distributions P and Q on the sample space, the KL divergence is defined as:

$$\mathrm{KL}(P, Q) = \sum_x P(x) \log\left(\frac{P(x)}{Q(x)}\right)$$

Intuitively, KL(P,Q) measures how difficult it is to distinguish P from Q based on observations. We will use the following consequence, known as Pinsker's inequality for any event A [4]:

$$2(P(A) - Q(A))^2 \leq \mathrm{KL}(P, Q),$$

We will also use the the fact that KL-divergence decomposes additively over independent samples. We consider 0-1 rewards and the following family of problem instances, with parameter $\varepsilon > 0$ to be adjusted in the analysis:

$$I_j = \begin{cases} \mu_i = \frac{1}{2} + \frac{\varepsilon}{2} & \text{for arm } i = j, \\ \mu_i = \frac{1}{2} & \text{for each arm } i \neq j, \end{cases} \tag{4.1}$$

Distinguishing between two Bernoulli distributions with means $\frac{1}{2}$ and $\frac{1}{2} + \epsilon$ requires $T = O(1/\epsilon^2)$ samples. This follows from a standard KL-divergence (or Pinsker) argument.

4.1.2 Flipping multiple coins: "best-arm identification"

We consider a multi-armed bandit problem with K arms, where each arm is a biased random coin with unknown mean $\mu_i \in [0, 1]$, and pulling arm i yields a reward $X_{t,i} \sim \mathrm{Bernoulli}(\mu_i)$ independently across arms and rounds; after T rounds, the algorithm outputs $y_T \in \{1, \ldots, K\}$ as its prediction of the optimal arm $i^* \in \arg\max_i \mu_i$, which is known as the best-arm identification problem. We are only concerned with the quality of prediction here.

Let the set of arms be $[K] := \{1, \ldots, K\}$, $\mu(a)$ denotes the mean reward of arm a, and a problem instance is specified as a tuple $I = (\mu(a) : a \in [K])$.

For concreteness, we say that an algorithm for best-arm identification is *good* if, for every problem instance I,

$$\Pr\big(y_T = a^*(I) \mid I\big) \geq 0.99 \tag{4.2}$$

For each problem instance I. We will use the family 4.1 of problem instances, with parameter $\epsilon > 0$, to argue that one needs $T \geq O\big(\frac{K}{\epsilon^2}\big)$ for any algorithm to "work", i.e., satisfy property 4.2, on all instances in this family.

Lemma 4.2. *Consider a best-arm identification problem with $T \leq \frac{cK}{\epsilon^2}$, for a small enough absolute constant $c > 0$. Fix any deterministic algorithm for this problem. Then there exist at least $\lceil K/3 \rceil$ arms a such that for the problem instances I_a defined in 4.1,*

$$\Pr[y_T = a \mid I_a] < \frac{3}{4}.$$

Corollary 4.3. *Assume T is as in 4.2. Fix any algorithm for "best-arm identification." Then*

$$\Pr[y_T \neq a] \geq \frac{1}{12},$$

where the probability is over the choice of arm a, the randomness in rewards, and any randomness in the algorithm (randomly selecting an arm gives a failure probability of at least 1/12). This follows directly from (4.2) as general case follows because any randomized algorithm can be expressed as a distribution over deterministic algorithms. We define another problem instance called the base instance:

$$I_0 = \{\mu_i = 1/2 \text{ for all } i\}.$$

Let T_a be the number of times arm a is played and let $Y_T \in [K]$ denote the arm output by the algorithm after T rounds (its estimate of the best arm). Under I_0, many arms are neglected:

$$\text{There are } \geq \frac{2K}{3} \text{ arms } j \text{ such that } \mathbb{E}_0[T_j] \leq \frac{3T}{K}. \tag{4.3}$$

$$\text{There are } \geq \frac{2K}{3} \text{ arms } j \text{ such that } P_0(y_T = j) \leq \frac{3}{K}. \tag{4.4}$$

By Markov's inequality we have:

$$\Pr\left(T_j \geq \frac{24T}{K}\right) \leq \frac{\mathbb{E}_0[T_j]}{24T/K}$$

$$\Pr\left(T_j \geq \frac{24T}{K}\right) \leq \frac{3T/K}{24T/K}$$

$$\Pr\left(T_j \geq \frac{24T}{K}\right) \leq \frac{1}{8}$$

By the complement event we have $\Pr\left(T_j \leq \frac{24T}{K}\right) > \frac{7}{8}.$

Let $A = \left\{j : \mathbb{E}_0(T_j) \leq \frac{3T}{K}\right\}$ and $B = \left\{j : P_0(y_T = j) \leq \frac{3}{K}\right\}$.

By (4.3) and (4.4), we have $|A| \geq \frac{2K}{3}$ and $|B| \geq \frac{2K}{3}$.
Since the sets of arms in (4.3) and (4.4) must overlap on least $K/3$ arms, we conclude:

$$|A \cap B| \geq |A| + |B| - K \geq \frac{2K}{3} + \frac{2K}{3} - K = \frac{K}{3}$$

Hence we have at least $K/3$ arms satisfying both events A and B:

$$\Pr\left(T_j \leq \frac{24T}{K}\right) \geq \frac{7}{8} \quad \text{and} \quad P_0(y_T = j) \leq \frac{3}{K}.$$

For each arm a, define the reward sequence sample space $\Omega_a^t = \{0,1\}^t$. The full sample space is $\Omega = \prod_{a \in [K]} \Omega_a^T$. Fix an arm j satisfying 4.3 and 4.4. We show

$$P_j(Y_T = j) \leq \frac{1}{2}. \tag{4.5}$$

Before proceeding, we determine the threshold K_0 for the case split. In Case 2, the KL-divergence argument requires that $P_0(y_T = j) \leq \frac{3}{K}$ is small enough to be dominated by the Pinsker bound. Specifically, we need

$$\frac{3}{K} \leq \frac{1}{8},$$

which gives $K \geq 24$. We therefore set $K_0 = 24$ and split the proof accordingly. Additionally, we choose the constant $c > 0$ small enough so that $\delta = \sqrt{\frac{1}{2} \cdot 48c} \leq \frac{1}{8}$, which gives $c \leq \frac{1}{192}$.

Proof. Case 1, $K \leq 24$:
Since K_0 is a fixed constant, we have

$$\sqrt{KT} = O(\sqrt{T}).$$

Thus, it suffices to prove a regret lower bound of order $O(\sqrt{T})$. For $K \leq 24$, a constant, the simpler two-arm KL-divergence argument already gives the desired bound. We now prove the lower bound in the case $K = 2$.

We construct a probability space describing all possible reward outcomes. For each arm $a \in [K]$ and pull index $t \in [T]$, let $r_t(a)$ be a Bernoulli random variable with $\mathbb{E}[r_t(a)] = \mu(a)$. All random variables $\{r_t(a)\}$ are assumed to be mutually independent. The collection $(r_t(a) : a \in [K], t \in [T])$ is called the *rewards table*, where $r_t(a)$ represents the reward received the t-th time the algorithm pulls arm a. The sample space is $\Omega = \{0,1\}^{K\times T}$, where each outcome $\omega \in \Omega$ corresponds to a realisation of the entire rewards table. Any event concerning the algorithm can be interpreted as a subset $A \subseteq \Omega$, consisting of all outcomes for which the event occurs. Each problem instance I_j induces a probability distribution P_j over Ω, defined by

$$P_j(A) = \Pr[A \mid \text{instance } I_j], \quad A \subseteq \Omega.$$

Assume for contradiction that the algorithm succeeds for both instances I_1 and I_2. Let $A = \{y_T = 1\} \subset \Omega$ be the event that the algorithm predicts arm 1. Then $P_1(A) \geq \frac{3}{4}, P_2(A) \leq \frac{1}{4}$, so $P_1(A) - P_2(A) \geq \frac{1}{2}$. To arrive at a contradiction, we use a KL-divergence argument:

$$2\big(P_1(A) - P_2(A)\big)^2 \leq \mathrm{KL}(P_1, P_2) \quad \text{(by Pinsker's inequality)}$$

$$\mathrm{KL}(P_1, P_2) = \sum_{a=1}^{K}\sum_{t=1}^{T} \mathrm{KL}\big(P_1^{a,t}, P_2^{a,t}\big) \quad \text{(by the chain rule)}$$

$$\mathrm{KL}(P_1, P_2) \leq 2T \cdot 2\epsilon^2.$$

Simplifying we get:

$$P_1(A) - P_2(A) \leq \epsilon\sqrt{2T} < \frac{1}{2} \quad \text{whenever} \quad T \leq \left(\frac{1}{4\epsilon}\right)^2.$$

If the number of rounds T is too small, the algorithm cannot reliably pick the best arm — the difference in probability of success between similar instances is bounded and too small to succeed.

Case 2, $K > 24$:
We use the KL-divergence construction. Set $\epsilon = \sqrt{cK/T}$ and define I_0 and I_j as above. Since each arm j is pulled at most $24T/K$ times we have $\mathrm{KL}(P_0^*, P_j^*) \leq m \cdot 2\epsilon^2 = \frac{24T}{K} \cdot 2 \cdot \frac{cK}{T} = 48c$. This gives $\mathrm{KL}(P_0^*, P_j^*) \leq C$ for some constant $C > 0$. From Pinsker's inequality we have $|P_0^*(A) - P_j^*(A)| \leq \sqrt{\frac{1}{2} \cdot \mathrm{KL}(P_0^*, P_j^*)} = \delta$.

We know $P_0^*(A) \leq 3/K$ from 4.4 and $P_0^*(A') \leq 1/8$ from the markov inequality.

For A: $P_j^*(A) \leq P_0^*(A) + \delta \leq \frac{3}{K} + \delta$
For A': $P_j^*(A') \leq P_0^*(A') + \delta \leq \frac{1}{8} + \delta$

Hence we have:

$$P_j^*(A) \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4} \quad \text{and} \quad P_j^*(A') \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

and:

$$P_j(Y_T = j) = P_j^*(A) + P_j^*(A') \leq \frac{1}{2}.$$

Thus, with constant probability, the algorithm fails to identify the optimal arm. Interpreting the arm selected at round t as the prediction of a best-arm identification algorithm, 4.3 implies

$$\Pr(a_t \neq a) \geq \frac{1}{12}.$$

Under instance I_a, the optimal arm has mean $1/2 + \epsilon$, while all other arms have mean $1/2$. Hence the instantaneous regret is

$$\Delta(a_t) = \mu^* - \mu(a_t) = \epsilon$$

whenever $a_t \neq a$. Therefore,

$$\mathbb{E}[\Delta(a_t)] = \Pr(a_t \neq a) \cdot \epsilon \geq \frac{\epsilon}{12}.$$

Summing over all rounds,

$$\mathbb{E}[R(T)] = \sum_{t=1}^{T} \mathbb{E}[\Delta(a_t)] \geq \frac{\epsilon T}{12}.$$

Finally, substituting $\epsilon = \sqrt{cK/T}$ gives

$$\mathbb{E}[R(T)] \geq O(\sqrt{KT}).$$

□

This lower bound demonstrates that the exploration–exploitation trade-off is fundamentally unavoidable in multi-armed bandit problems. In particular, any algorithm must collect enough samples to reliably distinguish optimal from suboptimal arms and this limitation results in the fact that no strategy can achieve sub-linear regret significantly better than $O(\sqrt{KT})$ in the worst case.

5 Numerical Example and Interpretations

Let us now consider a real problem instance to simulate our different bandit algorithms on. Suppose we are a supermarket chain and have 5 advertisements and we want to display the ad that gets the highest click-rate. We model this problem as a bandit problem instance where the advertisements correspond to the arms of a bandit instance and the rewards of the arms correspond to each ad's click-rate which we model using Bernoulli random variables i.e the average probability that a user's click on the advertisement. Let us assume the true click probabilities/rewards are specified by the mean reward vector $\mu = (0.1, 0.2, 0.3, 0.4, 0.5)$ with $K = 5$. Let us now numerically simulate this problem instance over a time horizon of a million rounds. We will plot the empirical regrets for each algorithm as well as the expected regret bound for each algorithm.

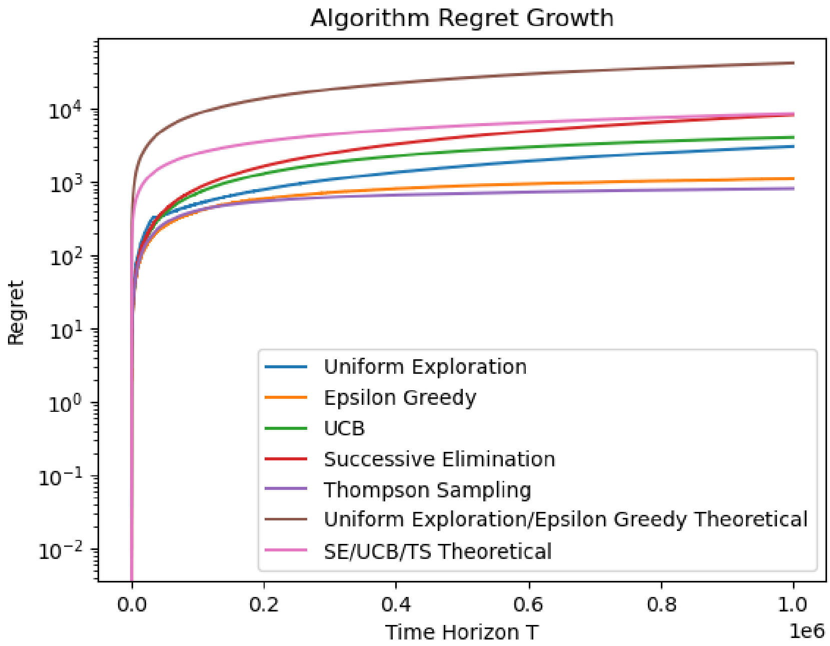

Figure 1: Cumulative regret over time.

From the regret plot as shown in Figure 1, there are two main points to take away from this simulation. First, each algorithm incurs less regret than their corresponding theoretical upper bounds. This is expected since these bounds are worst-case guarantees and are generally not tight for specific problem instances. The second observation is that the algorithms exhibit a clear ordering in performance, with Thompson Sampling achieving the lowest regret, followed by Successive Elimination, UCB and Epsilon-Greedy, while Uniform Exploration performs the worst. This ordering can be explained by the efficiency with which each algorithm allocates its samples. Uniform Exploration continues to sample all arms equally, including clearly suboptimal ones, resulting in many wasted pulls. Successive Elimination and UCB improves upon this by progressively discarding arms that appear suboptimal, and Epsilon-Greedy can exploit gaps and uses information gathered so far thereby concentrating more samples on promising arms. Thompson Sampling is even more efficient, allocating pulls according to the posterior probability that each arm is optimal. As a result, it focuses exploration on uncertain but potentially optimal arms while rapidly exploiting strong candidates. This increasingly efficient allocation of samples leads to progressively lower cumulative regret.

Overall, this simulation provides an empirical complement to the theoretical results established earlier. In particular, it illustrates that the regret bounds derived for each algorithm are not only theoretically meaningful but also reflect their practical performance. Moreover, the observed ordering of algorithms is consistent with their theoretical guarantees, with more adaptive methods achieving lower regret in practice.

6 Conclusion

Overall, in this report, we studied multi-armed bandit problems and analysed the performance of several classical algorithms, including Uniform Exploration, Successive Elimination, and Thompson Sampling. We derived regret upper bounds for each algorithm and showed that adaptive strategies achieve significantly improved performance compared to non-adaptive exploration. We further established a fundamental lower bound of order $O(\sqrt{KT})$ on the expected regret. Our numerical experiments supported the theoretical findings, illustrating both the relative performance of the algorithms and the gap between worst-case guarantees and typical instance behaviour. The framework analysed and established here provides a foundation for many real world applications such as in reinforcement learning, sequential portfolio selection, online advertising, dynamic pricing, etc. [5]

7 Limitations and Future Work

There are a few areas of this current multi-armed bandit model which can be extended and further studied.

First, the relaxation of the assumption that the rewards for these bandit problems need to be fixed over the whole time horizon. Instead rewards can change over time and therefore a different model is needed. This leads to research into non-stationary bandit models and algorithms based on discounted rewards or sliding-window estimates, which place greater emphasis on recent observations and can adapt to changing environments [6].

Furthermore, Our current bandit model assumes we can only make decisions solely on the basis of previous rewards. Further research can look into relaxing this restriction by exploring bandit problems where there is side information available before each decision called contextual bandits [7].

Finally, another area of further research is to study multi-armed bandits from a dynamic programming perspective, where the objective is to characterise optimal policies, for example via the Gittins index [8], rather than focusing solely on asymptotic regret guarantees.

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